3B06 Wave Nature of Light

3.6.1 · 3.6.2 Young's double-slit interference

Two coherent slits produce alternating bright and dark fringes on a distant screen. At small angles, spacing between adjacent fringes of the same type is proportional to wavelength and screen distance and inversely proportional to slit separation. Slit separation is not slit width, and bright-to-nearest-dark spacing is only half the fringe spacing.

$$\Delta y \approx \frac{\lambda D}{a}$$

Young's double slit: Δy: adjacent fringe spacing; λ: wavelength; D: slit-to-screen distance; a: slit separation. Same length unit; D ≫ a and small angles; coherent illumination.

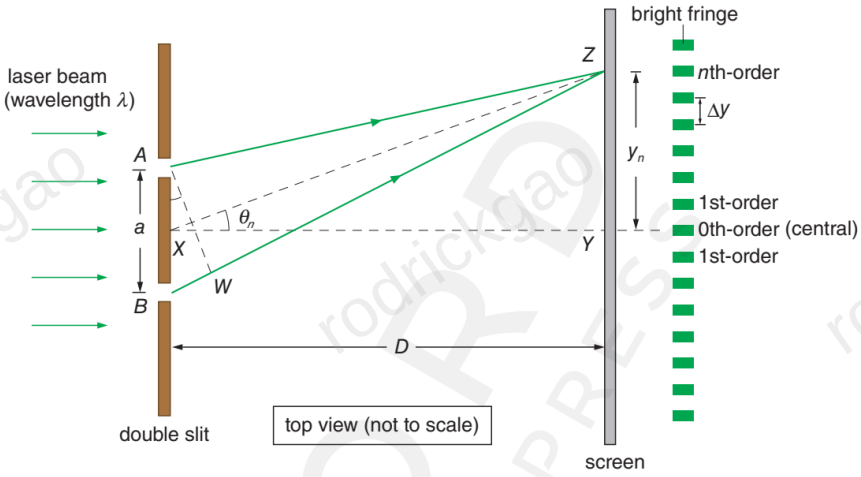


Fig 6.2i *Investigating the relationship between Δy, λ, D and a*

a is slit separation, D is screen distance and Δy is centre-to-centre fringe spacing. The small-angle approximation is required for Δy ≈ λD/a.

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3.6.3 · 3.6.4 · 3.6.5 Diffraction gratings and orders

3.6.3 · 3.6.4 Diffraction gratings

For normal incidence on equally spaced slits, a bright order forms when rays from adjacent slits have an integer-wavelength path difference. At a fixed order, longer wavelength or smaller slit separation increases the diffraction angle. Moving the screen changes positions but not angles. Convert line density to slit separation by taking its reciprocal with consistent units.

$$d\sin\theta = n\lambda$$

Grating with normal incidence: d: adjacent slit spacing (m); θ: angle from the straight-through direction; n: integer order; λ: wavelength (m).

3.6.5 Highest diffraction order

The grating equation and the condition that sine cannot exceed 1 determine the greatest allowed integer order. An order exactly at 90° travels along the grating plane and does not reach a screen in front, so exclude it when counting observable orders there. Total bright orders usually include the central zero order and symmetric orders on both sides.

$$n \leq \frac{d}{\lambda}$$

$$n_{\max} = \left\lfloor \frac{d}{\lambda} \right\rfloor$$

n: non-negative integer order; d: grating spacing; λ: wavelength, same units. Equality gives θ = 90°, a grazing order that does not reach a forward screen.

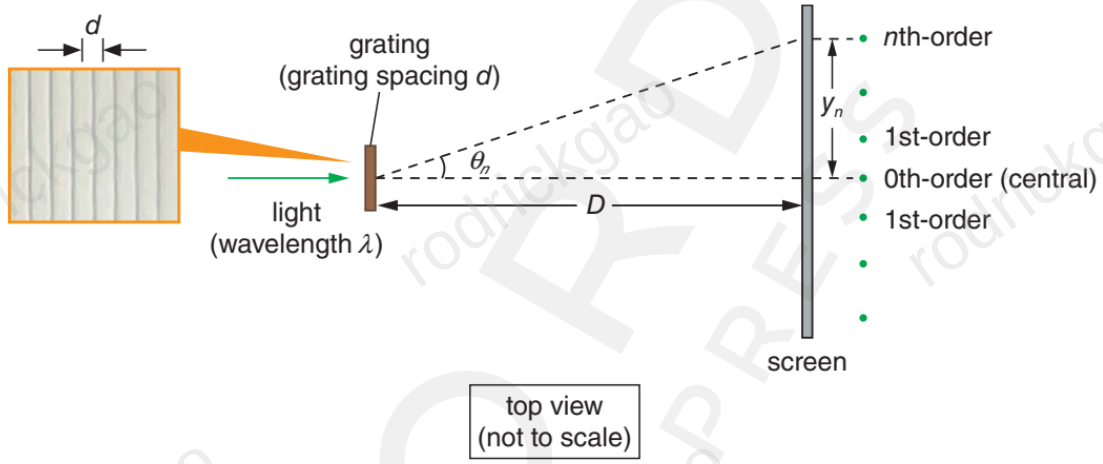


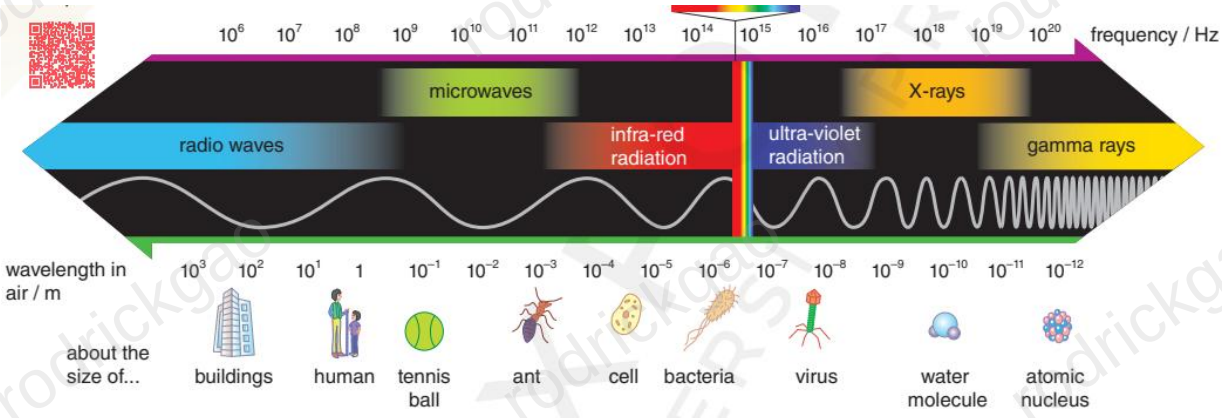
Fig 6.3i *Fringes formed when light passes through a grating.*

Measure θ from the central direction. d is adjacent slit spacing, not the grating width; changing D changes screen positions but not diffraction angles.

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3.6.6 Electromagnetic spectrum and applications

From longest to shortest wavelength, the spectrum runs through radio waves, microwaves, infrared, visible light, ultraviolet, X-rays and gamma rays. Frequency increases along this sequence because all travel in vacuum at $c = 3.0 \times 10^8\ \mathrm{m\,s^{-1}}$, with $c = f\lambda$. Typical uses include broadcasting, microwave heating and satellite communication, infrared imaging, ultraviolet sterilisation and X-ray imaging. Read the wavelength scale carefully; neighbouring bands have no perfectly sharp boundaries.



Wavelength decreases and frequency increases from left to right; typical length scales are shown below.

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3.6.7 Atmospheric transmission: microwaves and ultraviolet

Suitable microwave bands pass through the atmosphere and support satellite communication, whereas the atmosphere blocks much solar ultraviolet. Absorption therefore depends on wavelength. If an

atmospheric-transmission graph is supplied, read it at the specified wavelength rather than assuming every wavelength within a named band transmits equally well.

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